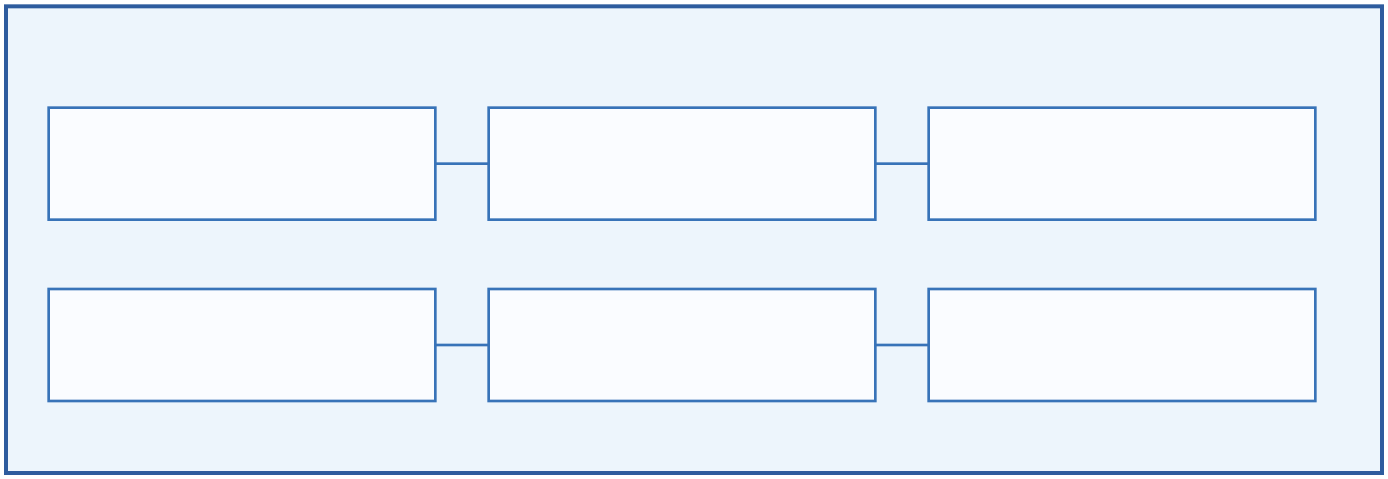
EXHIBIT A: STREAMING PUB/SUB -> DATAFLOW -> BIGQUERY MEDALLION TOPOLOGY

DOC_TYPE=TECHNICAL_EXHIBIT | SECTION=ARCH-MEDALLION-02 | ELEMENT=diagram | AMENDMENT=0 | EFFECTIVE=2026-02-01 | AMENDS

Architecture Specification ARCH-MEDALLION-02 (Lakehouse Medallion Pipeline):

1. Ingestion Layer: Global Cloud Pub/Sub topics with schema validation & dead-letter queues.
2. Stream Processing: Dataflow Streaming Engine (Apache Beam) performing windowing,
   PII tokenization via Cloud DLP, and Storage Write API exactly-once commits.
3. Medallion Storage in BigQuery + BigLake Iceberg:
   - Bronze Layer: Raw append-only change-data-capture (CDC) tables.
   - Silver Layer: Deduplicated, schema-enforced enterprise entity tables.
   - Gold Layer: Governed semantic marts & Vertex AI Feature Store embeddings.